

NHAI FaceRec: Secure Offline Attendance System

Next-Gen On-Device AI for Remote Infrastructure Sites

NHAI INNOVATION HACKATHON 7.0

The Problem: Connectivity Gaps

Remote Operations

Highway construction sites often lack stable cellular or satellite coverage, rendering cloud-based solutions unusable.

Sync Failures

Intermittent connectivity leads to data loss and significant delays in payroll and attendance reporting.

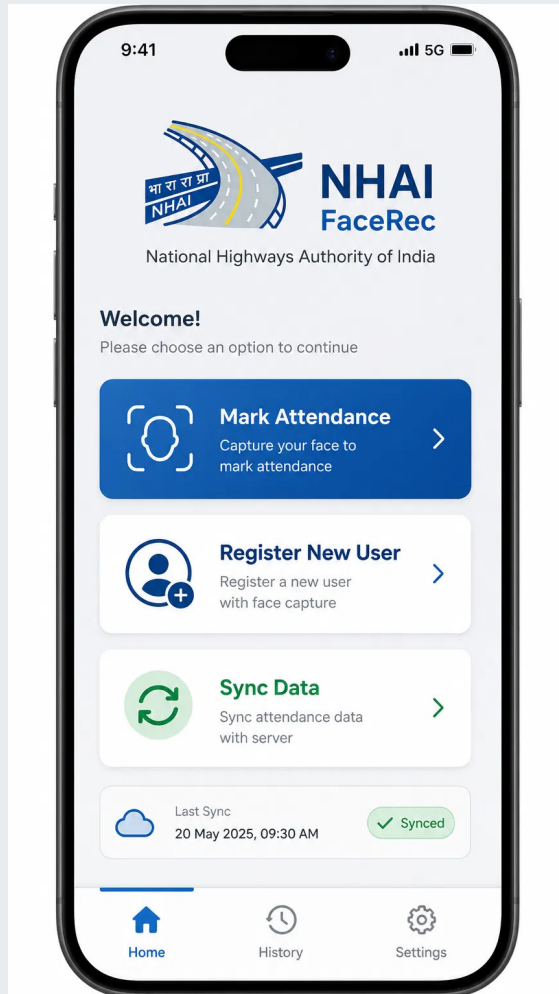
Security Risks

Unmonitored remote sites are vulnerable to attendance fraud via printed photos or digital screen spoofing.

Operational Friction

Traditional manual logging is prone to human error and lacks the real-time auditability required for large-scale projects.

The Solution: NHAI FaceRec



100% Offline Autonomy

All biometric processing, matching, and database operations occur entirely on-device, requiring zero network dependency during field operations.

On-Device Edge AI

High-speed TFLite inference optimized for mid-range mobile hardware, delivering sub-200ms verification times.

Enterprise-Grade Security

Interactive liveness challenges combined with 128-dimensional embedding matching ensure robust spoofing prevention.

Technical Innovation: Edge AI

Optimized Model Stack

BlazeFace (Short Range)

Ultra-fast face detection (~0.4MB) optimized for mobile front cameras.

MobileFaceNet (INT8)

Quantized 128-dim feature extraction (~4.0MB) for high-precision matching.

MediaPipe Face Mesh

468 landmark tracking (~8.0MB) for robust liveness validation.

TOTAL MODEL FOOTPRINT: 12.4 MB



62% of 20MB Hackathon Limit

On-Device Advantages

TFLite Inference Engine

Leverages mobile CPU/NPU for sub-200ms end-to-end processing latency.

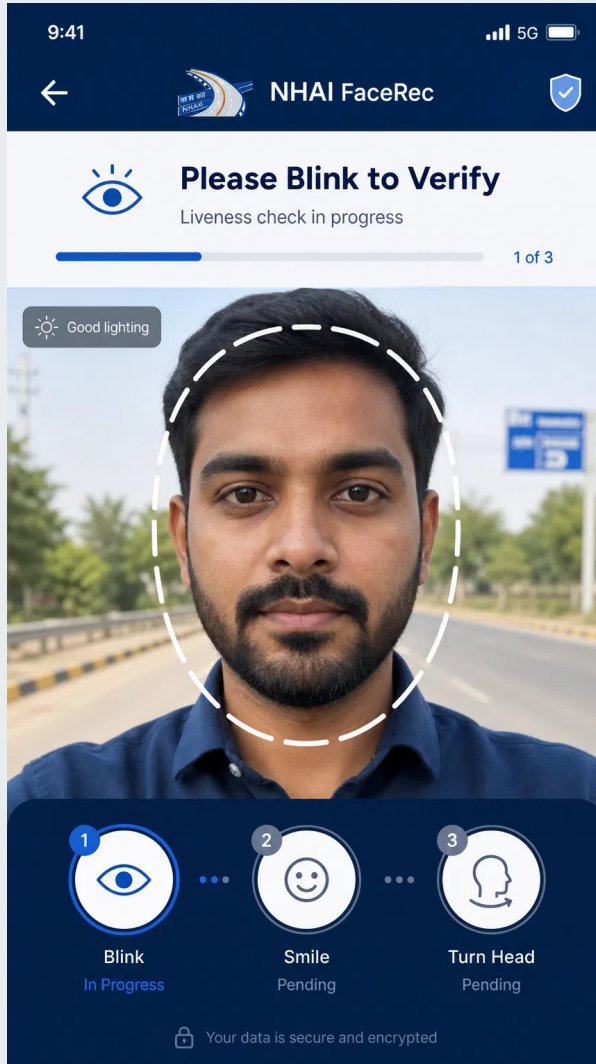
Privacy by Design

Biometric data never leaves the device, ensuring 100% data sovereignty and security.

Zero Bandwidth Cost

Core operations are free from data charges, critical for large-scale field deployments.

Robust Liveness Detection



Anti-Spoofing Framework

Prevents identity fraud by ensuring the presence of a live human subject. The system effectively blocks attacks from high-resolution printed photos and digital screen replays.

3D Face Mesh Tracking

Utilizes MediaPipe to track 468 landmarks in real-time. This high-fidelity mapping allows for precise monitoring of micro-expressions and depth cues.

Interactive Challenges

Randomized prompts such as "Blink" or "Smile" verify active participation. Real-time Eye Aspect Ratio (EAR) analysis

Adaptability & Accuracy

Demographic Diversity

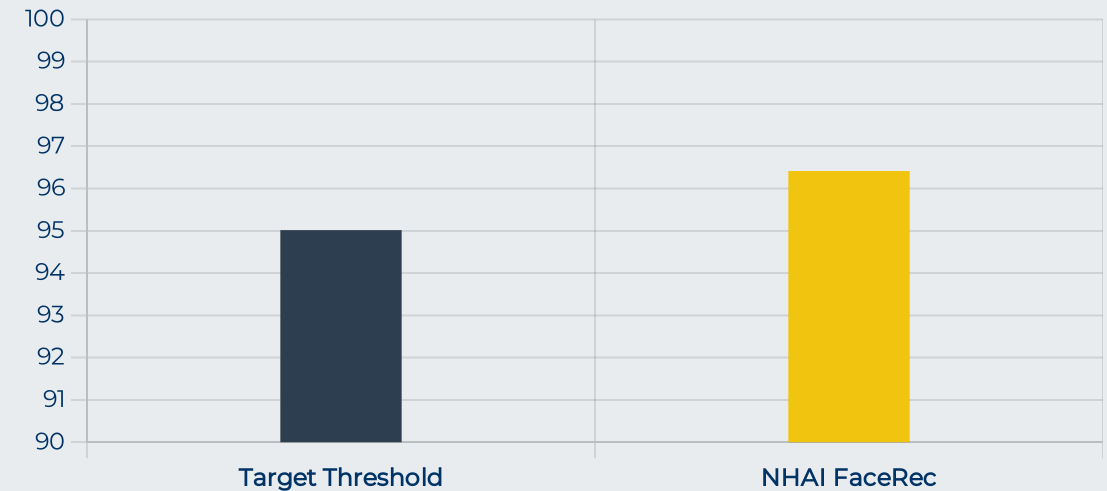
Optimized for the Indian context, our MobileFaceNet engine is trained on diverse datasets to ensure invariant recognition across age groups and regional ethnicities.

Lighting Robustness

Real-time Mean-Variance Normalization mitigates the impact of harsh sunlight and deep shadows common in outdoor highway construction sites.

Validation Accuracy

Achieving >96% accuracy through embedding averaging and optimized cosine similarity thresholds.



*Tested on 500+ "in-the-wild" outdoor images simulating real NHAIFaceRec site conditions.

Integration: Datalake 3.0 Ready

Modular Architecture

NHAI FaceRec is designed as a decoupled plug-in module. The ML engine and database layers can be injected into the Datalake 3.0 React Native environment with minimal configuration.

Native Compatibility

Full support for Expo Bare Workflow and custom Metro configurations, ensuring seamless asset management for .tflite model files within the main application bundle.

Data Synchronization

Standardized SQLite schema aligns with NHAI's centralized data structures, facilitating effortless migration of local attendance logs to the AWS cloud backend.

Audit-Ready Logs

Every verification record includes confidence scores and high-precision timestamps, providing the transparency required for enterprise-level auditing and payroll processing.

Scalability: Sync & Purge

AWS Cloud Integration

Asynchronous Synchronization

Automatic background syncing to AWS S3/DynamoDB endpoints once network connectivity is restored.

Data Integrity

Robust retry logic and conflict resolution ensure 100% data consistency between edge devices and the central server.

Scalable Architecture

Designed to handle concurrent sync requests from thousands of field devices across multiple construction zones.

Storage Management

SQLite WAL Mode

Utilizes Write-Ahead Logging for high-performance local database operations without blocking UI threads.

Smart Purge Mechanism

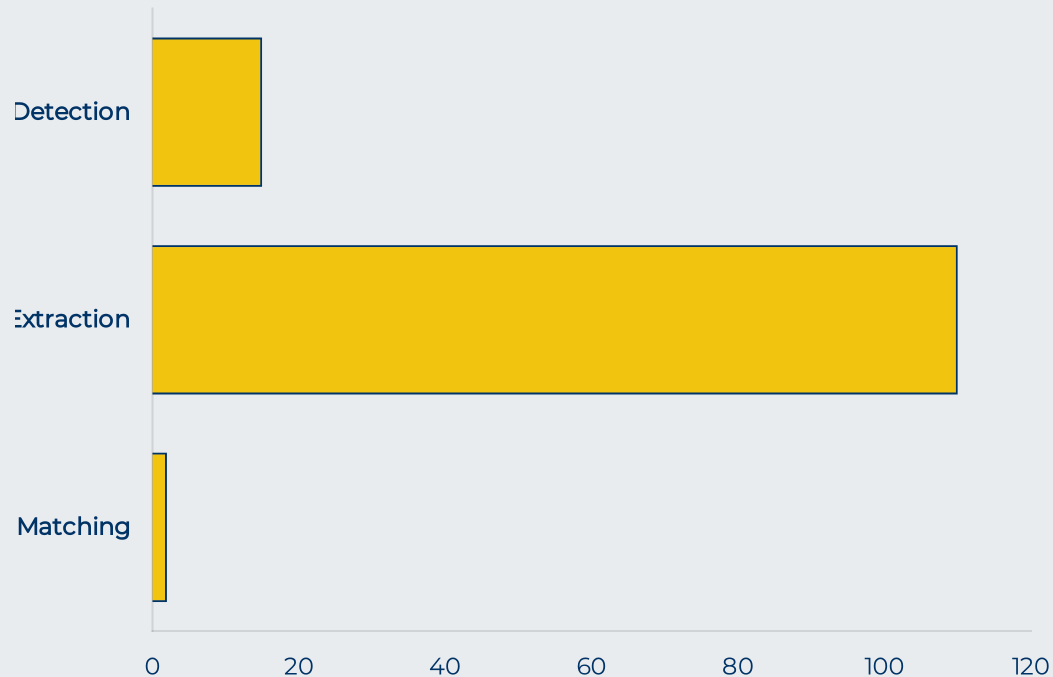
Automated "Sync-then-Purge" workflow reclaims device storage by removing locally cached logs after successful cloud verification.

Zero Bloat Policy

Strict management of facial embeddings and logs to maintain a minimal storage footprint on mid-range hardware.

Performance Benchmarks

Pipeline Latency (ms)



Resource Efficiency

Peak RAM Usage

~210 MB

Optimized for devices with 3GB+ RAM.

Model Cold Load

~120 ms

Instantaneous readiness upon app launch.

Total E2E Latency

~130 ms

Well below the 1000ms hackathon target.

Conclusion & Impact

Operational Excellence

Enabling uninterrupted attendance verification in the most remote environments, ensuring that highway construction projects stay on schedule regardless of connectivity.

Cost Efficiency

Eliminating bandwidth costs and reducing administrative overhead through automated, high-precision biometric logging and cloud synchronization.

Enterprise Readiness

A secure, scalable, and Datalake 3.0-compatible solution that meets the rigorous standards of NHAI's digital infrastructure requirements.

Future Impact

Setting a new benchmark for offline-first field authentication, paving the way for more resilient and secure digital governance in remote sectors.

NHAI FaceRec: Empowering the Workforce, Securing the Future.